

# Solution Requirements

## Phase 2: Requirement Analysis

shopEZ E-commerce Marketplace | MERN Stack | June 2026

### 1. Functional Requirements (FR)

The system must deliver the following capabilities, structured by user access level via Role-Based Access Control (RBAC).

#### FR-1: User and Authentication Services

| ID     | Requirement                                                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------------------------------|
| FR-1.1 | Standard users must be able to register an account with a unique email, name, and hashed password.                               |
| FR-1.2 | Users must be able to log in securely to retrieve a signed JSON Web Token (JWT) session.                                         |
| FR-1.3 | The system must restrict administrative dashboard routes to users holding the admin or seller roles.                             |
| FR-1.4 | Standard users must be able to manage an address book containing multiple shipping locations, selecting a default for checkouts. |

#### FR-2: Product Catalog and Discovery

| ID     | Requirement                                                                                                         |
|--------|---------------------------------------------------------------------------------------------------------------------|
| FR-2.1 | The database must list products displaying name, description, price, stock inventory, categories, and images.       |
| FR-2.2 | The frontend must provide full-text search with live autocomplete suggestions.                                      |
| FR-2.3 | The interface must support dynamic filters based on Category, Price Range, Minimum Ratings, and Stock Availability. |
| FR-2.4 | The frontend must support sorting: Price (Low to High, High to Low), Ratings, and Release Date (Newest).            |

### FR-3: Shopping Cart and Wishlist Lifecycle

| ID     | Requirement                                                                                              |
|--------|----------------------------------------------------------------------------------------------------------|
| FR-3.1 | The system must offer a persistent shopping cart linked to the user context.                             |
| FR-3.2 | Users must be able to modify item quantities in the cart, subject to real-time stock limits.             |
| FR-3.3 | The system must maintain a saved Wishlist, permitting users to move favorites to cart in a single click. |

### FR-4: Checkout, Coupons and Payments

| ID     | Requirement                                                                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-4.1 | The system must guide the customer through a multi-step checkout wizard (Shipping Address, Shipping Speed, Coupon Apply, Mock Payment, Order Confirmation). |
| FR-4.2 | The coupon engine must validate active coupon codes, applying percentage or fixed discounts and recalculating itemized taxes and shipping fees.             |

|        |                                                                                                      |
|--------|------------------------------------------------------------------------------------------------------|
| FR-4.3 | The checkout must communicate with a mock payment intent endpoint to securely complete transactions. |
|--------|------------------------------------------------------------------------------------------------------|

## FR-5: Administrative Operations

| ID     | Requirement                                                                                                                   |
|--------|-------------------------------------------------------------------------------------------------------------------------------|
| FR-5.1 | Admins and sellers must hold complete CRUD privileges over products.                                                          |
| FR-5.2 | Sellers and admins must be able to upload product images, stored locally via Multer.                                          |
| FR-5.3 | Admins must be able to modify order fulfillment milestones (Placed, Confirmed, Packed, Shipped, Out for Delivery, Delivered). |
| FR-5.4 | Admins must be able to block/unblock customer accounts, instantly revoking active sessions.                                   |
| FR-5.5 | Admins must be able to manage coupon campaigns (create, toggle active state, delete).                                         |

## 2. Non-Functional Requirements (NFR)

### NFR-1: Security and Protection

| ID      | Requirement                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------|
| NFR-1.1 | Passwords must be hashed using the Bcrypt algorithm (salt round factor 10) before database commits.              |
| NFR-1.2 | Session JSON Web Tokens must be transmitted via HTTP-Only, Secure cookies to prevent XSS-based token extraction. |
| NFR-1.3 | Enforce strict API middleware validation to deny unauthenticated access to protected routes.                     |

### NFR-2: Performance and Optimization

| ID      | Requirement                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------------------|
| NFR-2.1 | The UI must maintain high responsiveness utilizing React's virtual DOM to update components without full-page reloads. |
| NFR-2.2 | Product images must be served via Express static file serving from /backend/uploads for quick local access.            |
| NFR-2.3 | API responses for the catalog must complete within 200ms under standard loads.                                         |

### NFR-3: Reliability and Integrity

| ID      | Requirement                                                                                                                          |
|---------|--------------------------------------------------------------------------------------------------------------------------------------|
| NFR-3.1 | Database commits during checkout must ensure atomic stock deductions to prevent oversells.                                           |
| NFR-3.2 | If the primary MongoDB cluster is unavailable, the backend seeder must automatically instantiate a local In-Memory MongoDB fallback. |

### NFR-4: Usability and Accessibility

| ID      | Requirement                                                                                                |
|---------|------------------------------------------------------------------------------------------------------------|
| NFR-4.1 | The user interface must utilize a fully responsive layout that fits mobile, tablet, and desktop viewports. |
| NFR-4.2 | Design styling must utilize Outfit or Inter fonts with clear typography hierarchies.                       |